Math 32 Curriculum

Precalculus: Trigonometry | Middlesex School Mathematics | Resource: Math 32 Workbook

COURSE AT A GLANCE

Trigonometric Foundations

REQUIRED

pp. 3-6 Pythagorean Theorem; Special Right Triangles; pp. 29-33 Review Questions; pp. 7-12 Introduction to Radians; Related Angles; Radians and Degrees; pp. 13-14 Arc Length; pp. 15-21 Unit Circle Coordinates; Sine, Cosine, Tangent; pp. 24-28 Trig Functions Beyond Acute Angles; Non-Unit Circles; pp. 34-45 Multiple Angle Solutions; Reciprocal Functions; pp. 22-23 Right Triangle Applications

UNIT LEARNING OBJECTIVES

I can use the Pythagorean theorem to determine an unknown side of a right triangle.
I can determine exact side lengths in 30-60-90 and 45-45-90 triangles.
I can convert angle measures between degrees and radians.
I can determine coterminal and reference angles for a given angle.
I can calculate arc length from a central angle and radius.
I can determine exact coordinates on the unit circle.
I can evaluate all six trigonometric functions exactly for common angles.
I can solve right triangles using trigonometric ratios.
I can apply right-triangle trigonometry to contextual problems.
I can evaluate trigonometric functions from a point on a circle or terminal side.
I can solve a basic trigonometric equation for all angles in a specified interval.

Periodic Functions and Modeling

REQUIRED

pp. 46-49 Graphing Sine and Cosine; pp. 50-62 Vertical and Horizontal Transformations; pp. 63-68 Graphing All Transformations; pp. 69-76 Modeling Problems; pp. 77-85 Graphing Cosecant, Secant, and Tangent

UNIT LEARNING OBJECTIVES

I can graph basic sine and cosine functions using key features.
I can identify amplitude, midline, period, and phase shift from an equation or graph.
I can graph transformed sine and cosine functions.
I can write sine and cosine equations that represent a periodic graph.
I can construct a sinusoidal model from contextual information or data.
I can interpret the parameters and outputs of a sinusoidal model in context.
I can graph tangent, secant, and cosecant functions with their asymptotes.
I can determine whether a function is even, odd, or neither from its graph or formula.

Inverse Functions and Trigonometric Equations

REQUIRED

pp. 86-91 Calculating Angles; Inverse Trig Functions; pp. 92-98 Solving Trig Equations, Part 2; pp. 99-100 Compositions of Trig and Inverse Trig Functions

UNIT LEARNING OBJECTIVES

I can read inverse-trigonometric notation and distinguish it from reciprocal notation.
I can evaluate inverse sine, inverse cosine, and inverse tangent expressions.
I can explain how restricted domains make inverse trigonometric functions possible.
I can solve trigonometric equations exactly or approximately on a specified interval.
I can express all solutions of a trigonometric equation in general form.
I can evaluate compositions of trigonometric and inverse trigonometric functions.

Non-Right Triangles

REQUIRED

pp. 102-110 Law of Sines; Ambiguous Case; pp. 111-115 Law of Cosines; Mixed Practice

UNIT LEARNING OBJECTIVES

I can solve a non-right triangle using the Law of Sines.
I can determine the number of possible triangles in the ambiguous SSA case.
I can solve a non-right triangle using the Law of Cosines.
I can select and apply an appropriate method to solve a triangle in context.

Trigonometric Identities

REQUIRED

pp. 116, 119-123 Pythagorean, reciprocal/quotient, even/odd, cofunction, and required practice

UNIT LEARNING OBJECTIVES

I can rewrite trigonometric expressions using reciprocal and quotient identities, including `tan(theta) = sin(theta)/cos(theta)`.

I can rewrite trigonometric expressions using the Pythagorean identities.

I can use even/odd and cofunction identities to rewrite trigonometric expressions.

I can verify a trigonometric identity using the required identity families.

Additional Trigonometric Identities

EXTENSION

pp. 117-118, 124-126 Double-angle and sum/difference identities

UNIT LEARNING OBJECTIVES

I can apply double-angle identities.

I can apply half-angle identities.

I can apply angle sum and difference identities.

I can determine exact trigonometric values using extension identities.

Triangle Area from SAS

EXTENSION

pp. 101 New Area Formula

UNIT LEARNING OBJECTIVES

I can calculate the area of a triangle from two sides and the included angle.

Polar Coordinates

REQUIRED

pp. 127-128 Polar Coordinates and Coordinate Conversion

UNIT LEARNING OBJECTIVES

I can plot and identify points in polar coordinates.

I can represent a polar point using equivalent coordinate pairs.

I can convert points between polar and rectangular coordinates.

Polar Functions

EXTENSION

pp. 129-132 Polar Functions; pp. 133-135 Polar/Rectangular Function Conversion

UNIT LEARNING OBJECTIVES

I can graph and interpret basic polar functions.

I can convert equations between polar and rectangular forms.

Vectors

EXTENSION

pp. 136-140 Vector Concepts and Operations; pp. 141-144 Components, Magnitude, Direction, and 3D Vectors

UNIT LEARNING OBJECTIVES

I can represent a vector geometrically and in component form.

I can add, subtract, and multiply vectors by scalars.

I can determine vector magnitude, direction, and unit vector.

I can resolve a vector into horizontal and vertical components.

I can perform basic operations with vectors in three dimensions.

Parametric Equations

EXTENSION

pp. 145-147 Introduction to Parametrics

UNIT LEARNING OBJECTIVES

I can graph a plane curve defined by parametric equations with its orientation.

I can eliminate a parameter to obtain a rectangular equation.

I can create or interpret parametric equations for motion in the plane.